Research Cluster_Health Tech

MA I

Master's thesis proposal
Request of acceptance of a master's thesis topic

General information

First name	Matouš	Surname	Pešice
Student ID number	52006687	Date of birth	28.7.1998

Request / master's thesis topic (max. 280 characters)

I request the acceptance of the following thesis topic (Official Title):
Development of a Low-Latency ML-Based Motion Tracking System for Gamified Knee Rehabilitation in Virtual Reality.

Presentation of the problem/motivation

Current rehabilitation setups suffer from high latency (~200ms), causing "laggy" avatar behavior that breaks immersion and hinders therapy. Traditional knee exercises lack engagement, leading to poor patient compliance. This project addresses the technical bottleneck of latency while introducing gamification to improve therapy outcomes.

Aim and leading question

Aim: To create a real-time VR feedback loop where an avatar responds to skeletal tracking within <10ms.
Leading Question: How does a low-latency, goal-oriented "Architect" game influence the stability and engagement of patients during single-leg balance exercises compared to standard visual feedback?

Method and planned strategy

Hardware/Software Pipeline: Integration of a standard RGB camera with ML skeletal extraction to replace high-latency legacy systems..
Gamification Implementation: Developing a VR exercise prototype centered on proprioceptive training, using a goal-oriented "Architect" mechanic to shift focus from internal joint movement to external virtual objectives..
User Evaluation: A pilot study focusing on the "System Feel"—measuring how users perceive the responsiveness (latency) and their level of engagement compared to traditional exercises.

Roadmap:

Kick-off Meeting: 8.1.2026
Half-time meeting: 5.3.2026
Final meeting: 15.5.2026
Planned submission: 31.5.2026

NON-DISCLOSURE

I request preclusion from public access for the above mentioned master's thesis until _____ (please insert date – max. of 5 years starting with the submission date of the thesis) due to the following reasons:
*I hereby agree that the locked thesis will also be checked on plagiarism. If this is not okay for external companies, it is the student's obligation to proceed as described in the Academic Walkthrough chapter "3.1.7 Plagiarism".

ETHICS ASSESSMENT

Are Human subjects incolved in the thesis project?

- ☑ Yes
- ☐ No

SUPERVISOR (to be obtained by the student)

Prof. Bernhard Hollaus, PhD

Name

Bernhard Hollaus

Date, signature

Company, contact details (incl. email address)

With my signature, I agree that the supervised thesis will be checked with plagiarism detection software. This includes work which is precluded from public access.

EVALUATOR MCI (to be obtained by the student - only if supervisor is external)

Name

Date, signature

STUDENT

Matouš Pešice

Digitally signed by Matouš Pešice
Date: 2025.12.18 17:40:24 +01'00'

Date, signature student

APPROVAL

Daniel Sieber

cn=Prof. Dr. Dipl.-Ing. Daniel Sieber, o=MCI | The Entrepreneurial School ©, ou=Medical & Health Technologies, email=daniel.sieber@mci.edu, c=AT
2025.12.19 15:43:02 +01'00'

Date, signature head of study program